

CHESTER, CT, USA

WMO: 720545

Lat: 41.384N

Lon: 72.506W

Elev: 127

StdP: 99.81

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 00169

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.8	-10.9	-22.8	0.5	-10.8	-20.9	0.6	-7.9	10.2	-1.9	9.1	-2.3	3.7	330	0.460

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.9	31.3	23.4	29.9	22.7	28.0	21.6	25.3	29.0	24.4	27.6	23.6	26.4	3.1	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.0	19.2	26.6	23.0	18.1	25.3	22.7	17.7	25.0	78.0	28.4	74.2	28.0	71.4	26.4	28.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.5	7.5	6.5	DB	-17.1	34.0	3.1	2.1	-19.4	35.5	-21.2	36.8	-23.0	38.0	-25.3	39.5	(4)
(5)			WB	-18.0	26.4	3.0	1.1	-20.2	27.1	-21.9	27.8	-23.6	28.4	-25.7	29.2	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	11.1	-1.7	0.2	3.9	9.6	15.4	19.7	23.4	22.5	18.3	12.5	6.6	2.3		
	DBStd	9.47	5.26	5.16	4.69	4.20	4.10	3.38	2.77	2.71	3.70	4.42	4.48	5.02		
	HDD10.0	1282	363	277	197	55	4	0	0	0	0	24	119	242		
	HDD18.3	3058	621	509	447	263	111	23	2	2	47	186	351	497		
	CDD10.0	1692	0	2	8	44	171	292	416	387	248	103	18	4		
	CDD18.3	429	0	0	0	2	19	66	159	130	46	7	0	0		
	CDH23.3	2884	0	0	0	21	161	440	1178	840	227	18	0	0		
	CDH26.7	783	0	0	0	4	42	107	370	211	47	2	0	0		
(14) Wind	WSAvg	2.8	3.2	3.4	3.5	3.2	2.7	2.5	2.3	2.2	2.3	2.7	3.0	3.0		
(15) Precipitation	PrecAvg	1214	90	79	107	102	93	105	94	110	113	109	100	112		
	PrecMax	1516	174	214	238	212	186	259	237	248	260	389	168	201		
	PrecMin	888	47	39	16	32	20	12	48	47	29	17	30	35		
	PrecStd	172	35	34	51	46	38	65	37	56	60	78	34	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.8	16.0	20.9	27.0	31.1	32.3	33.9	32.5	31.1	25.9	19.0	16.2		
		MCWB	12.8	11.9	15.2	17.4	21.5	22.8	25.2	25.1	24.3	18.9	15.0	15.0		
	2%	DB	11.2	12.3	17.0	22.1	27.4	29.2	32.1	30.2	27.7	22.7	17.3	13.7		
		MCWB	9.9	10.1	11.2	14.1	19.4	21.3	23.9	23.6	21.5	19.4	14.4	12.7		
	5%	DB	7.9	10.0	12.9	18.6	24.8	27.5	30.1	28.8	26.1	21.2	15.9	12.4		
		MCWB	6.4	7.8	8.8	12.6	17.6	20.3	22.9	22.4	20.7	18.5	13.0	11.6		
	10%	DB	6.1	7.4	11.1	17.0	22.4	26.1	28.7	27.6	23.9	19.0	13.7	10.3		
		MCWB	3.9	4.7	7.2	10.7	16.0	19.4	22.5	21.8	20.3	16.7	11.1	8.9		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	13.8	15.8	17.7	22.5	24.7	26.7	26.5	25.1	22.3	17.5	16.0		
		MCDB	13.7	14.7	19.0	21.9	28.5	28.5	31.8	30.2	29.1	24.3	18.1	16.0		
	2%	WB	9.9	10.5	12.3	16.0	20.9	23.3	25.3	25.3	23.6	20.6	16.0	13.0		
		MCDB	10.2	11.6	14.3	20.0	25.1	27.1	29.3	28.7	25.8	21.6	16.8	13.2		
	5%	WB	7.1	7.8	9.5	14.3	19.5	22.3	24.5	24.4	22.7	19.0	13.6	11.9		
		MCDB	7.8	9.4	13.4	17.1	23.1	25.8	28.1	27.0	24.7	20.6	14.7	12.4		
	10%	WB	4.0	5.0	7.6	12.5	18.0	21.2	23.7	23.5	22.0	17.4	11.7	8.8		
		MCDB	6.0	7.3	9.8	15.1	20.5	24.0	26.8	25.9	23.7	18.3	13.3	10.0		
(35) Mean Daily Temperature Range	MDBR		7.3	7.7	7.8	9.5	9.7	9.3	8.9	8.7	9.2	8.7	7.8	6.8		
	5% DB	MCDBR	9.3	10.0	10.4	12.1	13.0	10.7	10.2	9.6	9.8	9.3	9.3	8.7		
		MCWBR	9.5	8.1	6.5	6.6	6.6	5.3	4.3	4.3	5.2	5.7	7.6	8.8		
	5% WB	MCDBR	9.6	10.2	9.7	9.8	11.2	9.4	8.9	8.1	7.9	8.0	8.8	8.9		
		MCWBR	10.5	9.3	7.4	6.8	6.5	5.1	4.3	4.1	5.0	6.1	8.6	9.8		
(40) Clear-Sky Solar Irradiance	taub		0.300	0.309	0.331	0.368	0.414	0.455	0.475	0.445	0.377	0.338	0.310	0.298		
	taud		2.494	2.475	2.464	2.386	2.282	2.204	2.159	2.234	2.402	2.524	2.558	2.555		
	Ebn at Noon		859	906	921	903	864	826	804	818	859	860	843	835		
	Edh at Noon		74	87	99	115	131	142	147	133	105	83	69	65		
(44) All-Sky Solar Radiation	RadAvg		1.87	2.73	3.71	4.83	5.39	5.86	5.97	5.32	4.38	2.94	2.05	1.55		
	RadStd		0.15	0.24	0.29	0.52	0.54	0.59	0.33	0.30	0.34	0.28	0.16	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days					
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(47)	Regional (77 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+57		

Nomenclature: See separate page